

ORIGINAL INVESTIGATIONS

Association of *TSR1* Variants and Spontaneous Coronary Artery Dissection



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ABSTRACT

BACKGROUND Spontaneous coronary artery dissection (SCAD) is defined as a splitting of the coronary artery wall exclusive of iatrogenesis or trauma. Since the last decades, our knowledge of the diagnosis and prognosis and therapy for SCAD has advanced; however, its causes remain unknown.

OBJECTIVES This study sought to identify genes associated with SCAD development in the Chinese Han population.

METHODS Between November 2011 and January 2018, the authors enrolled 85 SCAD cases and 296 non-SCAD controls from the Chinese Han population. All 381 subjects enrolled underwent detection with whole exome sequencing, followed by Sanger sequencing for confirmation. Principle component analysis was used to evaluate the structure of the population. Haploview was used to analyze the linkage disequilibrium statistics of the variants. The author used 2 gene-based association tests, optimal sequence kernel association test and mixed effects score test, to identify the causal genes or variants of SCAD. Immunohistochemistry was used to detect the expression of *TSR1* in coronary artery tissues.

RESULTS Four genes with a suggestive association with SCAD ($p < 5.41 \times 10^{-5}$ in both the optimal sequence kernel association and mixed effects score tests) were identified, and *TSR1* was the top hit. All *TSR1* germline variants were either highly conserved across distinct species or lead to premature termination of protein syntheses. Furthermore, the expression of *TSR1* was detectable in human coronary artery tissues.

CONCLUSIONS This study describes the clinical characteristics of the Chinese Han population with SCAD and identified *TSR1* as a potential causal gene, which might bring about a further progress in diagnosis and treatment of the disorder. (J Am Coll Cardiol 2019;74:167-76) © 2019 by the American College of Cardiology Foundation.

Spontaneous coronary artery dissection (SCAD), as an increasingly important cause of acute coronary artery syndrome (ACS), is defined as a splitting of the coronary artery wall exclusive of iatrogenesis or trauma (1). For decades, the condition has been deemed quite rare, but recently, the identification of SCAD is increasing presumably on account of the heightened awareness as well as the greater use



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ABBREVIATIONS AND ACRONYMS

ACS	= acute coronary artery syndrome
AD	= aortic dissection
CAD	= coronary artery disease
DCM	= dilated cardiomyopathy
ExAC	= Exome Aggregation Consortium
FMD	= fibromuscular dysplasia
IHC	= immunohistochemistry
IVUS	= intravascular ultrasonography
LAD	= left anterior descending coronary artery
MAF	= minor allele frequency
MI	= myocardial infarction
MIST	= mixed effects score test
OCT	= optical coherence tomography
SCAD	= spontaneous coronary artery dissection
SKAT-O	= optimal sequence kernel association test
WES	= whole exome sequencing

of coronary artery angiography in patients with ACS and advanced intracoronary imaging, for example, optical coherence tomography (OCT) and intravascular ultrasonography (IVUS). According to recent studies from Australia and Japan (2,3), the incidence of SCAD ranges between 1.7% and 4% of ACS cases. Furthermore, SCAD accounted for 23% to 36% of ACS in younger women (2,4-6). The presentation of SCAD spans a wide range of clinical presentations and severities; however, patients experienced primarily acute myocardial infarction (MI) with distinct proportions of ST-segment elevation MI across available case series (2,5,7-10). Even worse, ventricular arrhythmias or sudden cardiac death were presented in 3% to 11% of SCAD cases, which was probably underestimated due to failure to identify individuals who died before the initial evaluation (5,7-9,11). In addition, SCAD challenges percutaneous coronary intervention in multiple ways, such as, the risk of iatrogenic catheter-induced dissection and stent malapposition after resorption of intramural hematoma.

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In the last decades, case reports and cohort studies have advanced our knowledge of precipitating and potential predisposing factors, as well as diagnosis, therapies, and prognosis for SCAD (12). Despite the advances in our understanding, the pathogenesis of SCAD, especially inheritance and genetics, is not well studied (13,14). Fibromuscular dysplasia (FMD) is the first condition observed associated with SCAD and dominates the described arteriopathies in the pathogenesis of SCAD. Since the first description in 2005 (15), subsequent studies have confirmed the association between SCAD and extracoronary FMD, where FMD accounts for 11% to 86% of cases in patients with SCAD (14). In addition to FMD, the potential pathogenesis of SCAD is punctuated by case studies that have proposed multiple contributions from pregnancy and hormone influences as well as systemic inflammatory conditions (13). As for genetics and inheritance, SCAD has been recognized as a heritable disorder (16). Mutations in many genes that are related to arterial fragility and dissection have been reported in SCAD cases, for example, mutations in *COL3A1* (cause of Vascular Ehlers-Danlos syndrome), mutations in *FBN1* (cause of Marfan syndrome), and mutations in *TGFBR1*, *TGFBR2*, and *SMAD3*, which are members of the TGF- β pathway gene set (cause of

Loeys-Dietz syndrome) (17-20). More recently, our group reported the gene profile and novel pathogenic genes associated with sporadic aortic dissection (AD), which included genes that are associated with the TGF- β pathway, collagens synthesis, and matrix metalloproteinases synthesis (21-23).

Nevertheless, solid evidence is required regarding whether the known connective tissue disorders are the cause of SCAD or may be occurring together with SCAD by chance in the same individuals. Besides the obscure association with these known genetic conditions, SCAD has not been conclusively demonstrated to be associated with any other genes or variants, either, due to insufficient cases discovered and its sporadic propensity (13). Identification of risk loci will contribute to our understanding of SCAD pathogenesis as well as diagnosis, treatment, and even prediction of prognosis. Thus, we performed whole exome sequencing (WES) to identify genes associated with SCAD development in the Chinese Han population.

METHODS

STUDY SUBJECTS AND HUMAN CORONARY ARTERY TISSUE. We enrolled 85 cases of SCAD (intimal dissection and intramural hematoma) determined by coronary angiography, IVUS, or OCT from Tongji Hospital, Tongji Medical College in Wuhan, China, between November 2011 and January 2018, as well as 296 non-SCAD control patients (Online Table 1) from communities of Wuhan and patients without SCAD in Tongji Hospital. Iatrogenic dissections or those arising from blunt trauma were excluded. Baseline characteristics were extracted from a centralized hospital database. All subjects were of Han Chinese origin based on family name and self-report. To replicate the results of gene-based association tests, we enrolled an independent cohort consisting of 53 cases and 201 controls from Tongji hospital, Beijing Fuwai Hospital, Wuhan Union Hospital, and Wuhan Asia Heart Hospital. Coronary artery tissues were collected from 3 victims who died of coronary artery disease (CAD), AD, and severe infection, respectively. The study was approved by the Tongji Hospital Review Board and all participants provided written informed consent before inclusion.

WHOLE EXOME SEQUENCING. Genomic DNA was isolated from peripheral blood leukocytes using a QIAamp DNA Mini Kit (Qiagen, Hilden, Germany). All genomic DNA samples were of high quality with an optical density 260/280 ratio ranging from 1.8 to 2.0. Of the 381 subjects, 344 (61 cases of SCAD and 283 control patients) were analyzed using an Illumina HiSeq X sequencer or Novaseq sequencer and the

other 37 (24 cases of SCAD and 13 control patients) were analyzed using an Ion Proton sequencer (Thermo Fisher Scientific, San Jose, California) ([Online Appendix](#)).

BIOINFORMATICS ANALYSIS. The principal component analysis for population stratification and genetic relationship matrix of GCTA (Genome-wide Complex Trait Analysis) were conducted with Eigenstrata ([24](#)) using the linkage disequilibrium-prune variants with a minor allele frequency (MAF) of >0.05 .

The qualified variants were annotated with ANNOVAR ([25](#)). The population allele frequency of a variant is based on data from the 1000 Genomes project, National Heart, Lung, and Blood Institute Grand Opportunity Exome Sequencing Project (ESP6500), and The Exome Aggregation Consortium (ExAC). We defined rare variants as variants with an MAF of <0.01 in the ESP6500 and the East Asian Population databases in the 1000 Genomes Project and ExAC. Deleterious variants were defined as: 1) loss-of-function alterations (nonsense variants, frameshift indels, and splicing); and 2) missense variants with either The Rare Exome Variant Ensemble Learner (REVEL) scores of >0.4 or The Mendelian Clinically Applicable Pathogenicity (M-CAP) scores of >0.025 ([26,27](#)).

GENE-BASED ASSOCIATION TEST. Of all qualified variants, we filtered for rare deleterious variants, and only genes with at least 2 rare deleterious variants in the combined population (85 cases and 296 control patients) were considered for the 2 downstream gene-based association tests, the optimal sequence kernel association test (SKAT-O) ([28](#)) and the mixed effects score test (MiST) ([29](#)), to evaluate the association between genes and SCAD risk.

PERMUTATION TEST OF GENE-BASED ASSOCIATION TEST. For correction of multiple testing, an experimental-wide significance threshold of 1,000 times permutations of phenotype labels was computed, which created the empirical distribution of minimal p values for all genes across permutations. Finally, a suggestive threshold at $\alpha = 0.05$ of the p value is 5.41×10^{-5} .

SANGER SEQUENCING. Sanger sequencing was used to confirm all rare damaging variants identified by SKAT-O and MiST tests and to replicate the result in an additional independent population with the primers listed in [Online Table 2](#).

HOMOLOGY MODELING. Structure modeling of mutant *TSR1* was carried out with SWISS-MODEL ([30](#)). The *Homo sapiens* crystal structure (PDB code: 6G18) ([31](#)) was used as a reference for mutant *TSR1*.

IMMUNOHISTOCHEMISTRY. Paraffin-embedded human coronary artery tissues were cut into 5- μ m sections, and immunohistochemistry (IHC) assays were performed as previously described ([32](#)). Antibody against *TSR1* (ABclonal, Wuhan, China) was used as the primary antibody.

CANDIDATE ANALYSIS AND WATERFALL PLOT.

Twenty-seven genes ([Online Table 3](#)) putatively associated with aortic dissection were considered candidate genes in the further analysis. A summary waterfall plot of the variants in genes associated with aortic dissection was generated using the R package “maftools” (R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

THE CLINICAL CHARACTERISTICS OF SCAD CASES.

From December 2011 to March 2018, we recruited 85 SCAD patients diagnosed with coronary angiography and who underwent dedicated intracoronary imaging (OCT and IVUS), in case of ambiguity. Clinical characteristics are summarized in [Table 1](#). Patients were 26 to 85 years of age (mean 55.0 years) and included 15 women (17.6%). Of the women, 2 patients (2.4%), who both delivered by caesarean section, had pregnancy-associated SCAD, with 1 during pregnancy and 1 at 9 days postpartum. At the time of SCAD diagnosis, 84 patients (98.8%) presented with ACS, with 48 patients (57.8%) with unstable angina, 27 (32.1%) with ST-segment elevation MI, and the rest (10.8%) with non-ST-segment elevation MI, and 1 patient was identified during electrocardiogram screening, followed by coronary artery angiography. In addition, 2 patients had ventricular tachycardia and ventricular fibrillation, respectively. Notably, the mean left ventricular ejection fraction, aortic dimension, left ventricular internal dimension, and left atrial internal dimension were 53.3%, 31.5 mm, 50.9 mm, and 37.1 mm, respectively. Abnormal left ventricular internal dimension or left atrial internal dimension was observed in more than one-half of SCAD cases, whereas enlargement of the aortic dimension was observed in nearly 25% of the participants ([Online Table 4](#)). Almost one-half of the patients had left ventricular wall motion abnormality ([Table 1](#)).

Across all individuals, 3 presented with SCAD >1 coronary artery. Of the 92 dissections, 48 (52.2%) occurred in the left anterior descending coronary artery (LAD), the most common coronary artery territory dissected, followed by the right coronary artery (40.2%). Specifically, the distribution of coronary artery segment dissections is presented in [Online Figure 1](#).

TABLE 1 Clinical Characteristics of the SCAD Cases (n = 85)

Age, yrs	55.0 ± 13.6
Female	15 (17.6)
Active tobacco use	32 (37.6)
Hypertension	44 (51.8)
Diabetes mellitus	24 (28.2)
Dyslipidemia	29 (34.1)
Cerebrovascular disease	7 (8.2)
Thyroid disease	4 (4.7)
Pregnancy-associated SCAD	2 (2.4)
Previous MI	16 (18.8)
Acute coronary syndrome	84 (98.8)
UA	48 (57.1)
STEMI	27 (32.1)
Non-STEMI	9 (10.8)
Normal ECG	2 (22.2)
Nonspecific ST-T-wave changes	2 (22.2)
T-wave inversions	4 (44.4)
ST-segment depression	1 (11.2)
VT/VF	2 (4.1)
Ejection fraction, %	53.3 ± 13.0
Ejection fraction <50%	25 (29.4)
AOD, mm	31.5 ± 3.7
LVID, mm	50.9 ± 6.7
LAID, mm	37.1 ± 6.5
IVS, mm	10.1 ± 1.6
LVPW, mm	9.8 ± 1.1
Left ventricular wall motion abnormality	38 (44.7)
SCAD involving >1 coronary artery	3 (3.5)
Coronary artery territory dissected	92
Left main artery	2 (2.2)
Left anterior descending artery	48 (52.2)
Circumflex artery	5 (5.4)
Right coronary artery	37 (40.2)
Successful PCI procedure	47 (55.3)
Aspirin	79 (92.9)
P2Y ₁₂ antagonists	76 (89.4)
Statins	78 (91.8)
ACE inhibitors/ARBs	57 (67.1)
β-Blockers	58 (68.2)

Values are mean ± SD or n (%), unless otherwise indicated.

ACE = angiotensin-converting enzyme; AOD = aortic dimension; ARB = angiotensin II receptor blocker; ECG = electrocardiogram; IVS = interventricular septum; LAID = left atrial internal dimension; LVID = left ventricular internal dimension; LVPW = left ventricular posterior wall; MI = myocardial infarction; NSTEMI = non-ST-segment elevation myocardial infarction; PCI = percutaneous coronary intervention; SCAD = spontaneous coronary artery dissection; STEMI = ST-segment elevation myocardial infarction; UA = unstable angina; VF = ventricular fibrillation; VT = ventricular tachycardia.

GENE-LEVEL ANALYSIS IDENTIFIED THE ASSOCIATION OF *TSR1* DELETERIOUS VARIANTS WITH SCAD.

We further performed WES of leukocyte-derived DNA in 85 SCAD cases and 296 control patients, and achieved an average coverage of 119-fold on target and 97.9% of bases covered at $\geq 20\times$ (Online Table 5). After extensive quality control (Online Figure 2), WES analysis revealed a total of 224,518 variants called across all samples in exonic regions, including 7,643 indels and 216,875 single-nucleotide variants.

The PCA revealed a homogenous population structure between the cases and control patients (Online Figure 3). According to the result of GCTA-GRM analysis, 1 individual from the control group was removed from the study group. For identification of potentially causative variants or genes associated with SCAD, a relatively inclusive maximum MAF of 0.01 was selected as the threshold to highlight the genes of interest, based on our limited data from the current study and published reports of SCAD prevalence in East Asians (3) and the genetic architecture (16). As such, we screened for variants with an MAF of <0.01 on the basis of the databases of the 1000 Genomes project and ExAC as well as ESP6500. We identified 46,206 variants harbored in 13,223 genes after filtration for the presence of deleterious variants (damaging missense, splicing site, nonsense, and frameshift indel variants).

Due to the limited power of the single-variant test in the WES study, we implemented gene-based association tests instead, with all deleterious rare variants filtered from the study (MAF of <0.01), to identify potentially causative genes of SCAD. The association between the genes and SCAD was evaluated using SKAT-O and MiST, both of which are linear combinations of unidirectional and variance component tests. Four genes showed a suggestive association with SCAD ($p < 5.41 \times 10^{-5}$ in both SKAT-O and MiST) (Online Table 6, Online Figure 4). The *TSR1* gene from the chromosome 17p13.3 region was the top hit according to the rank based on the number of net patients carrying the rare deleterious variants (SKAT-O $p = 6.03 \times 10^{-6}$; MiST $p = 1.72 \times 10^{-6}$) (Online Table 6). In total, 6 rare deleterious variants, consisting of 1 frameshift deletion, 1 nonsense, and 4 damaging missense variants, were observed in 8 independent SCAD cases. The characteristics of the subjects with multiple deleterious variants of *TSR1* are shown in Table 2. In view of the fact that the current studies prefer to exclude SCADs accompanied by atherosclerotic CAD, we filtered for patients without coronary atherosclerosis. Re-evaluation of associations between SCAD and genes in 5 cases (DP8211, XJ28703, DP5994, XJ29938, and SD40) and 295 control patients revealed that the gene *TSR1* was still the top hit (SKAT-O $p = 4.48 \times 10^{-5}$; MiST $p = 0.00$), along with the gene *ESRP2* (SKAT-O $p = 0.00$; MiST $p = 0.00$), both of which harbored 2 variants in 2 cases and in no control patients. Of the carriers of *TSR1* damaging variants, 3 presented with troponin-positive acute ST-segment elevation MI, including 1 young man 26 years of age (XJ28703) and 1 young woman at 9 days postpartum (DP8211). Further examination of their

TABLE 2 Multiple Rare Deleterious Variants of *TSR1* Identified in SCAD

Subject	Age at Diagnosis, yrs	Sex	Variant	Protein Alteration	Exon	MAF in Controls (n = 296)	MAF in 1,000 Genomes (EAS)	MAF in ExAC (EAS)	MAF in ESP6500
DP10984	74	Male	G2315A	R772Q	15	0	ND	0.0014	ND
DP5018	67	Male	G2315A	R772Q	15	0	ND	0.0014	ND
XJ20617	61	Male	G2315A	R772Q	15	0	ND	0.0014	ND
DP8211	34	Female	C1864T	R622C	11	0	0.001	0.0007	ND
XJ28703	26	Male	G1668A	W556X	10	0	ND	ND	ND
DP09432	57	Male	G1496C	R499P	8	0	ND	ND	ND
DP10389	49	Male	G1490A	R497Q	8	0	0.002	0.0027	0.0038
DC2264	50	Male	1_8del	M1fs	1	0	ND	ND	ND

EAS = East Asian; MAF = minor allele frequency; ND = not detected; SCAD = spontaneous coronary artery dissection.

parents revealed that both the putative causal variants originated from their fathers ([Online Figure 5](#)). The individual DP8211 (family1: III.1) presented as MI due to a distal dissection in LAD and her grandmother (family1: I.2) died of abdominal aortic aneurism. The individual XJ28703 had dissections in the left main and proximal left circumflex coronary arteries, and the electrocardiogram of his father (family2: I.1) showed ST-T-wave changes in leads II, III, and AVF, and Q waves in III and AVF ([Online Figure 5](#)). Considering the false-positive rate in WES, we confirmed all the rare deleterious variants by Sanger sequencing ([Figure 1](#)). In addition, we performed Sanger sequencing in 213 patients with CAD and 507 subjects with dilated cardiomyopathy (DCM), and the 6 variants were absent in both populations. The characteristics of these variants are shown in [Figure 1 and Table 2](#). Interestingly, except for the 2 premature truncating variants, the rest of the missense variants (n = 4) all lead to the alteration of arginine to another neutral amino acid. Multiple sequence alignment revealed the highly evolutionary conservativeness of the 4 residues where the missense variants occurred, and the residues lost after the nonsense variant ([Online Figure 6](#)).

We further sequenced all the exons of *TSR1* in an independent population, including 53 SCAD patients and 201 healthy control patients, and yielded 2 additional rare deleterious variants, 1 of which is the variant c.C1864T:p. R622C, the same variant harbored in individual DP8211 with no variant found in controls (SKAT-O p = 2.6×10^{-5} ; MiST p = 3.3×10^{-2}).

Following the identification of variants in *TSR1* that had potential functional association with SCAD, we continued to study their expression in coronary artery tissues based on the Genotype-Tissue Expression project (GTEx) database. Both coronary artery and aortic artery tissues had detectable expression of *TSR1*. As for the protein levels, IHC analysis revealed

that the expression of *TSR1* was still detectable in the LAD arteries from 3 victims who died of AD, CAD, and severe infection, respectively ([Online Figure 7](#)).

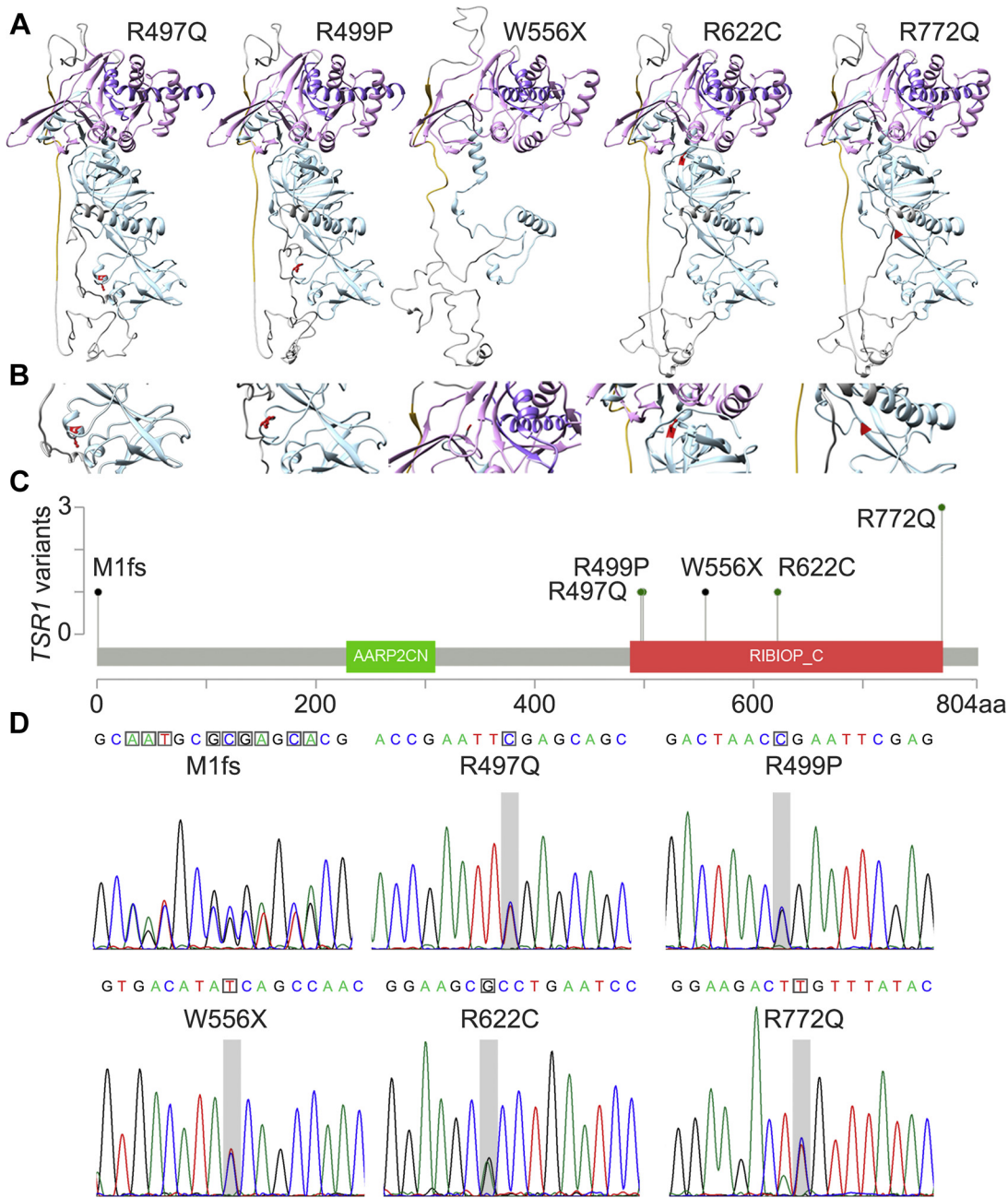
FURTHER GENE-LEVEL ANALYSIS OF GENES ASSOCIATED WITH AD. In addition to *TSR1*, we further explored variants harbored in genes putatively associated with AD; we observed 2 non-frameshift deletions, 2 non-frameshift insertions, 3 frameshift insertions, 2 nonsense and 227 missense variants in 380 samples ([Figure 2](#)). The ClinVar database was used to screen for pathogenic variants reported previously, but none were identified. After filtration for rare deleterious variants (MAF of <0.01), 20 genes, namely *COL4A5*, *FLNA*, *SLC2A10*, *FBN1*, *TGFB3*, *COL4A1*, *MFAP5*, *MYH11*, *EFEMP2*, *CSTF2T*, *PRKG1*, *NOTCH1*, *ELN*, *PLOD3*, *LOX*, *MYLK*, *TGFBR2*, *COL3A1*, *PLOD1*, and *SKI* ([Online Table 7](#)) were further investigated by gene-based association tests. However, none of the genes showed a suggestive association with SCAD (p value <0.05 in both tests) ([Online Table 8](#)).

DISCUSSION

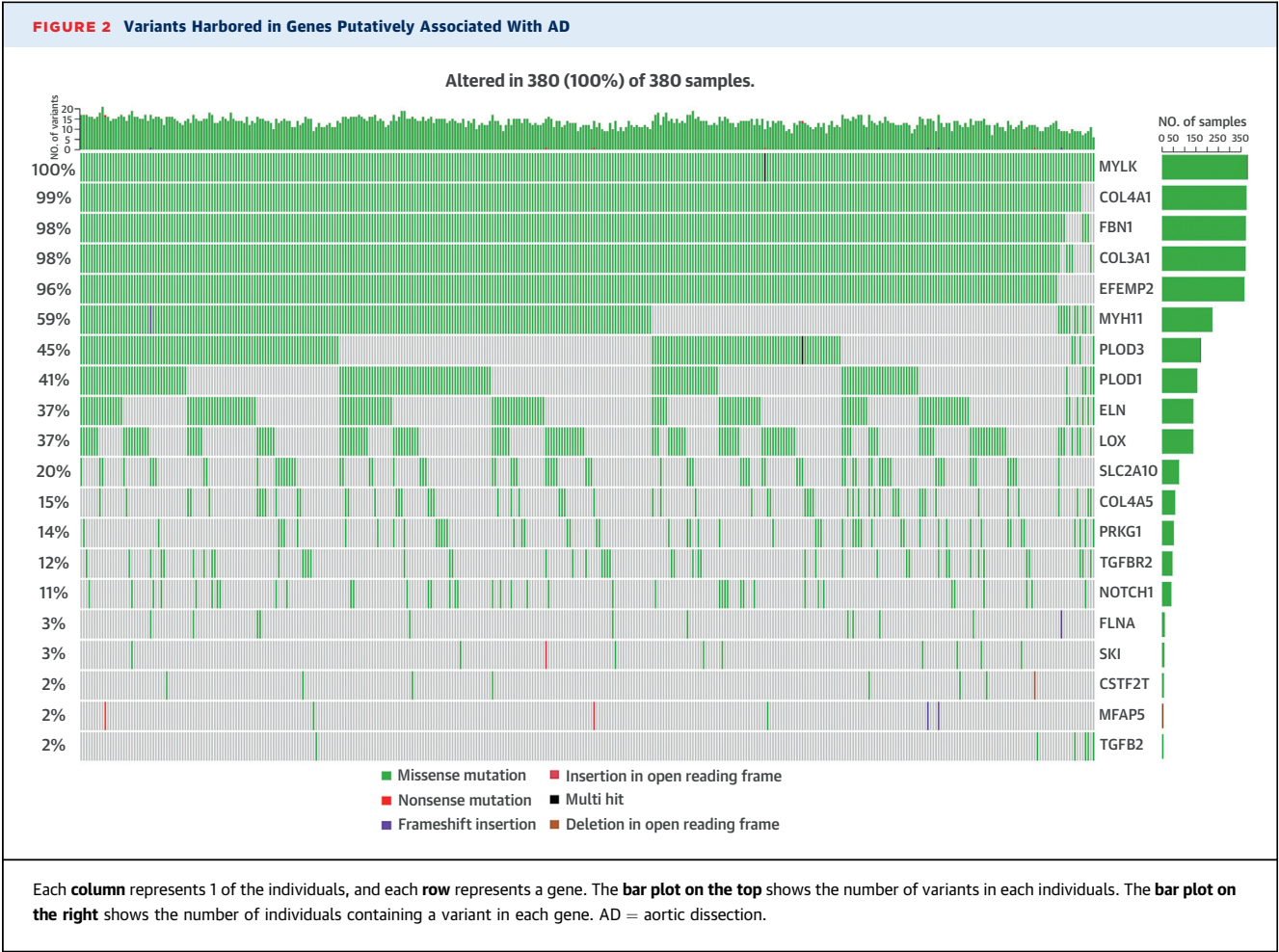
This first WES study of SCAD evaluated 381 subjects using gene-based association tests and revealed the association of *TSR1*, which maps to the chromosome 17p13.3 region, and SCAD genetic susceptibility ([Central Illustration](#)). All variants identified in the present study are either located in highly conserved regions or yield premature stop codons or frameshift mutations, indicating that these alterations may contribute to the dysfunction of *TSR1*. Further IHC analysis confirmed the detectable expression of *TSR1* in coronary artery tissues from humans, which suggested its potential role in SCAD development.

SCAD is described as a spontaneous dissection of the coronary artery wall without any iatrogenic cause or blunt trauma, whereas most current studies have focused on SCAD without atherosclerosis ([1](#)). Distinct from current studies, we did not rule out

FIGURE 1 Multiple Rare Deleterious Variants at *TSR1* Identified in SCAD Cases



The schematic structures **(A)** and details **(B)** of the multiple deleterious variants were predicted with SWISS-MODEL. The residues where the 4 missense and 1 nonsense variants occurred were highlighted **(red)**. **(C)** The mutations needle plot (created with MutationMapper) of the *TSR1* deleterious variants at the protein level across the spontaneous coronary artery dissection (SCAD) patients. A premature stop or a frameshift deletion is symbolized by a **black circle**, whereas a **green circle** denotes an altered amino acid. **(D)** All 6 *TSR1* variants identified in the current study were validated by Sanger sequencing.

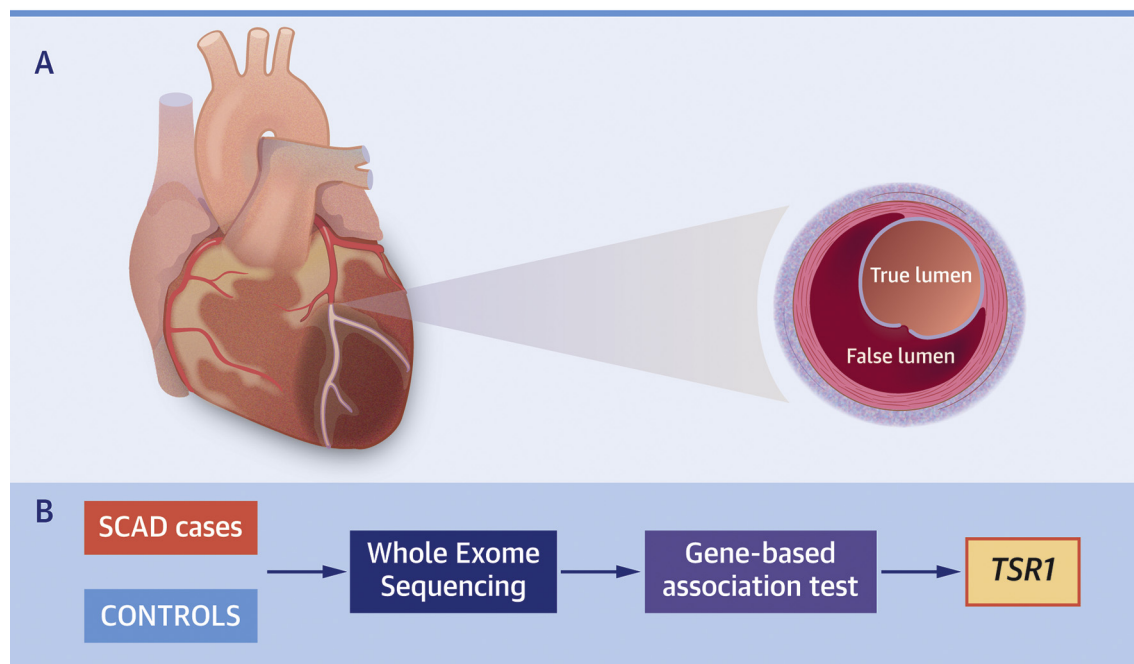


SCAD cases accompanied with atherosclerosis in the coronary artery because of our assumption that SCAD might be a genetic disorder with atherosclerosis being a precipitating factor or coexisting by chance. And it is interesting that the result of the analysis restricted to those with no atherosclerosis was similar to the result with atherosclerotic cases included. Currently, most studies have reported a strong female predominance (female sex and pregnancy), whereas in our study, females only accounted for a small fraction in SCAD cases, which might be explained by the distinct inclusion criteria and the genetic variance between the Gaussian population and the East Asian population.

With more and more SCAD cases being revealed due to applications of OCT and IVUS in coronary angiography and rising awareness, heritability has been documented more and more frequently; however, most are reported as case reports. Of the case reports, inherited arteriopathies and connective tissue disorders dominate the underlying cause of SCAD, such as vascular Ehlers-Danlos syndrome,

Marfan syndrome, and Loeys-Dietz syndrome (17-20). Nevertheless, these disorders only explain <5% of SCAD cases, and all these genes have not been demonstrated by more comprehensive studies (8,11,33). Of the limited cohort studies, the Mayo Clinic SCAD Registry identified 5 familial cases of SCAD, the first published reports suggesting a genetic predisposition of SCAD (16). Soon after that, the first medical genetic evaluation was performed on 59 SCAD cases from the registry and found a minor association between genes causal to connective tissue disorders and SCAD development, which is consistent with our results.

In the present study, not only did we describe the clinical characteristics of SCAD cases from the Chinese Han population, but we also performed WES on sporadic SCAD cases and non-SCAD control patients based on a case control approach, considering the scarcity of family history. In this study, a large proportion of SCAD cases merely presented as unstable angina, which is distinct from many studies. As we proposed, the reason might be a distinct population or a different

CENTRAL ILLUSTRATION Whole Exome Sequencing Identifies *TSR1* Associated With Risk of Spontaneous Coronary Artery Dissection

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Coronary artery dissection is one of the causes of acute myocardial infarction (A). Whole exome sequencing (WES) followed by gene-based association test identified the gene *TSR1* associated with spontaneous coronary artery dissection (SCAD) (B).

inclusion criterion. On the other hand, according to the current study, we might infer a continuous spectrum of clinical presentation of SCAD, which might explain the rarity of family history on account of healing and obscure presentation.

Despite the superiority of WES in identifying causal variants directly relative to genome-wide association study, the approach requires more powerful statistics than conventional methods due to limited power in detection of each rare variant's effect individually. As such, to obtain sufficient power for identification of rare causal loci, we considered SKAT-O (28) and MiST (29), both of which are a linear combination of unidirectional and variance-component tests with highest mean power across simulated datasets from the Moutsianas et al. (34) study. In view of the fact that the prevalence of SCAD is underestimated, and the scarcity of family history that we believe is due to low penetrance, the tests were performed based on variants with an MAF of <0.01 (commonly used MAF of rare variants) and those predicted to be deleterious by REVEL (28) and M-CAP (27) simultaneously.

Of all SCAD cases, a relatively dilated left ventricle was observed in almost one-half of the patients. Meanwhile, all patients with SCAD presented with CAD, most with acute coronary syndrome. Thus, to discriminate the distinctions between SCAD and non-SCAD control patients rather than between DCM and non-DCM patients or between CAD and non-CAD subjects, we further examined *TSR1* in 2 more populations consisting of DCM and CHD cases, respectively. The detection confirmed the unique enrichment of patients carrying *TSR1* germline variants in SCAD cases.

TSR1, one of the late-acting assembly factors that contributes to a sequential 40S subunit assembly, is characterized by 4 domains packing together to form a chalice-like structure (31,35). The domain I located at the N-terminal has a small GTPase-like fold, whereas the second, third, and fourth domains (II to IV) are β -barrels (35). In addition to the frameshift deletion variant that disrupted the whole gene and the nonsense variant that prematurely truncated the protein in the middle of domain II, the remaining altered residues caused by the 4 rare deleterious variants are located in

highly conserved regions of domains II, III, and IV, which indicated the relatively important role of these loci. *TSR1* is predicted to be a blocker that excludes binding with pre-40S particles of the GTPase eIF5B and the ATPase Rio1, and 60S subunit joining during the cytoplasmic maturation of the 40S ribosomal subunits (35). The surface of *TSR1* has several clusters of lysine and arginine residues that are mainly positively charged. Interestingly, in the present study, the alterations of *TSR1* from 4 missense variants are all arginine changing into another neutral amino acid that cause the dysfunction of *TSR1*. Moreover, *TSR1* is expressed in all tissues at the RNA level and in most tissues at the protein level, which was confirmed in the current study with IHC analysis of the coronary artery tissues. In light of the comprehensive expression of *TSR1*, we presume that the relatively dilated left ventricle and enlarged aortic arch might be a series of abnormalities caused by *TSR1* variants.

Because both AD and SCAD are disorders caused by artery dissection, we were wondering whether the mutations or mutant genes putatively associated with AD are related to SCAD, either. Interestingly, the gene-based association test identified no genes significantly relevant, indicating that SCAD is not the same as AD in genetics. Thus, in practice, genetics screening focusing on AD genes might be used cautiously. On the other hand, to jump to the conclusion that AD and SCAD have nothing in common in genetics is not wise, either, because the power of this study may not be enough to detect the potential associations.

STUDY LIMITATIONS. In view of the early stages of the studies on the genetics underlying SCAD, some limitations existed in the current study. FMD was not screened thoroughly in all SCAD cases because our study shed light on the identification of causal genes. The cases in this study constituted the largest population for which we could obtain sufficient DNA for WES and validation, but it is still relatively small for WES approaches. The population structure is distinct from most modern studies because the SCAD cases accompanied by atherosclerosis were not excluded from the current study. Meanwhile, limited by lack of family history and the small sample size, only a few family members of the index SCAD cases had genetic

detection. In addition, the present study lacks in vivo and in vitro experiments for further confirmation of the function of *TSR1*. For now, we have focused on the gene-based association tests and bioinformatics analysis, but more detailed in vitro and in vivo experiments are in progress.

CONCLUSIONS

The study described a profile of clinical characteristics in the Chinese Han population with SCAD. The WES identified a novel gene, *TSR1*, located in the chromosome 17p13.3 region, as a potential causal gene of SCAD. The detectable expression of *TSR1* in the coronary artery was confirmed by IHC analysis. Identification of the association between SCAD and *TSR1* might bring about a further progress in diagnosis and treatment of the disease.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: *TSR1* may play a causal role in SCAD.

TRANSLATIONAL OUTLOOK: Further studies are required to confirm the role of *TSR1* in SCAD development and its roles, both as a diagnostic marker of individuals at risk and as a target for therapeutic intervention.

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KEY WORDS case-control study, spontaneous coronary artery dissection, whole exome sequencing

APPENDIX For an expanded Methods section as well as supplemental figures and tables, please see the online version of this paper.